

MATH 213 – Exam Review Solutions

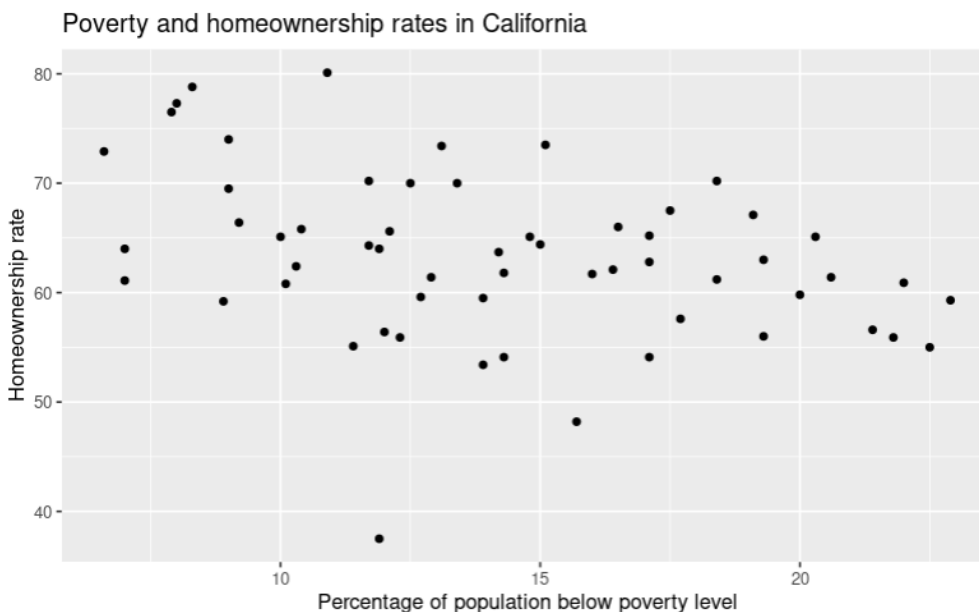
I have a dataset that contains information about all 58 counties in California. The data were gathered between 2006-2010. Here is a part of the code book for this dataset.

name	County name
metro	Whether the county contains a metropolitan area, with levels NO, YES
poverty	Percent of population in the county below poverty level (2006-2010)
homeownership	Home ownership rate for the county (2006 – 2010)

and here are the first few rows of the dataframe:

name	metro	poverty (%)	homeownership (%)
Alameda County	YES	11.4	55.1
Alpine County	NO	13.1	73.4
Amador County	NO	8.0	77.3

Here is a graph of “homeownership” versus “poverty”



- a. What are the cases in this study? (Think about what each dot in the scatterplot represents.)
The cases in this study are the counties in California.

- b. Write a paragraph describing the scatterplot in context. Start with a WWW sentence, and address the shape, strength, and direction of the association between the two variables. Throughout, speak in context and include proper units. Remember GEE – General trend, Examples, Exceptions. Follow our earlier examples to see good sentences you can use!

This graph displays the relationship between the homeownership rate and the percentage of the population below the poverty level for each of the 58 counties in California. Generally, there is a negative linear association between these variables, meaning that counties with lower percentages of their populations below the poverty level tend to have higher homeownership rates and vice versa. For example, counties that have less than 10% of their population below the poverty line tend to have homeownership rates of between 60% and 80%, whereas counties that have more than 20% of their population below the poverty level tend to have homeownership rates of between 55% and 65%. The relationship is moderately strong, meaning that there is some variation in homeownership rates for counties with similar rates of poverty. For example, counties that have about 12% of their population below the poverty line tend to have anywhere from 55% to 70% as their homeownership rate. There is a one county that is a notable exception to the trend in the plot that has about 12% of its population below the poverty level, but also has a very low homeownership rate of about 37%, much lower than the other counties of the same poverty level that have between 55%-75% of their population owning homes.

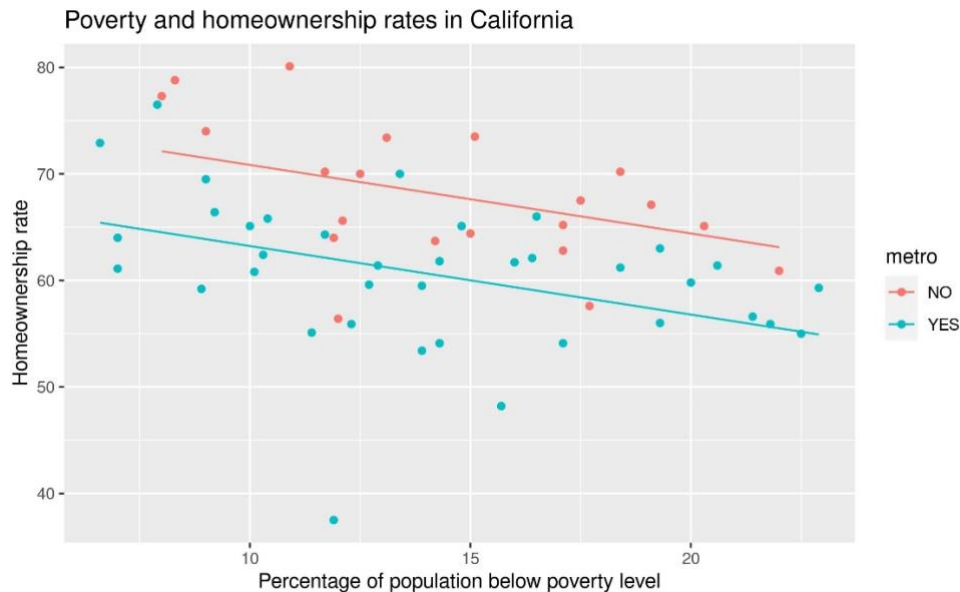
I had R create a model for predicting “homeownership” using “poverty” and “metro” as predictor variables. Here is the model summary output and a scatterplot that includes the model predictions.

```
Call:
lm(formula = homeownership ~ poverty + metro, data = county_complete2)

Residuals:
    Min       1Q   Median       3Q      Max
-24.5043  -3.4089   0.7394   4.1270  11.9210

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  77.2778     2.9983  25.774 < 2e-16 ***
poverty      -0.6437     0.1855  -3.469  0.00102 **
metroYES     -7.6137     1.6913  -4.502 3.54e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.186 on 55 degrees of freedom
Multiple R-squared:  0.3621,    Adjusted R-squared:  0.3389
F-statistic: 15.61 on 2 and 55 DF,  p-value: 4.268e-06
```



- c. Write the formula for the regression equation with a “hat” in the appropriate place.

$$\widehat{homeownership} = 77.2778 - 0.6437 * poverty - 7.6137 * metroYES$$

- d. Los Angeles County contains a metro area and has a poverty rate of 15.7%. What does the model predict for the homeownership rate in Los Angeles County?

The model predicts a homeownership rate of about 60% for Los Angeles County. My calculations are below:

$$\widehat{homeownership} = 77.2778 - 0.6437 * (15.7) - 7.6137 * (1) = 59.55801$$

- e. The actual homeownership rate in Los Angeles County is 48.2%. What is the residual for this county?

The residual for this county is

$$Observed - Predicted = 48.2\% - 59.56\% = -11.36\%.$$

This means LA county has a lower homeownership rate than its poverty rate would predict, meaning its point will be below the regression line.

- f. Write a sentence of interpretation for each of the three coefficients for the regression model. Make sure you understand the context of the data (what are the cases again?) before you begin to write,

and use that context in your sentences. Your sentences should include numbers and appropriate units of measurement.

Sentence 1:

77.2778: For a California county that has 0% of its population living below the poverty level and does not contain a metro area, the predicted homeownership rate is 77.28%.

Sentence 2:

−0.6437: When comparing only California counties that DO contain a metropolitan area or only California counties that DO NOT contain a metropolitan area, for each additional percent of the population living below the poverty level, the predicted homeownership rate decreases by 0.64 of a percentage point.

Sentence 3:

−7.6137: Comparing California counties with the same poverty rate, counties that contain a metropolitan area tend to have a predicted homeownership rate that is 7.6 percentage points lower than counties that do not contain a metropolitan area.

Hypothesis testing: Beer and Mosquitoes

In a study to investigate the effect of drinking beer on mosquito attraction, 43 men were randomly assigned to drink either a liter of beer or a liter of water, with 25 being assigned to the beer group and 18 to the water group. The number of mosquitoes approaching each man was then recorded, and is available in StatKey (Randomization Hypothesis Test for Difference in Means). We can see that the mean number of mosquitoes for the beer group was 23.6 while for the water group it is 19.222. We wish to see whether these data provide evidence that mosquitoes are attracted to beer drinkers (or, more precisely, that the mean number of mosquitoes approaching a beer drinker is significantly larger than the mean number of mosquitoes approaching a water drinker.)

1. Create some notation for these two sample means, and write down the number for the difference of these means.

$$\bar{x}_b - \bar{x}_w = 23.6 - 19.22 = 4.38$$

2. Create related notation for what we're trying to estimate: the difference of population means.

$$\mu_b - \mu_w$$

Note this quantity isn't equal to anything because the whole point is that we're trying to get a handle on this quantity by using the sample difference above.

3. State the null hypothesis for this test. Remember the null hypothesis is an equation involving the population parameters, and the alternate is an inequality. I've started it for you, though if you've used different notation than I did, you should change this to match your variable names.

$$H_0: \mu_b - \mu_w = 0$$

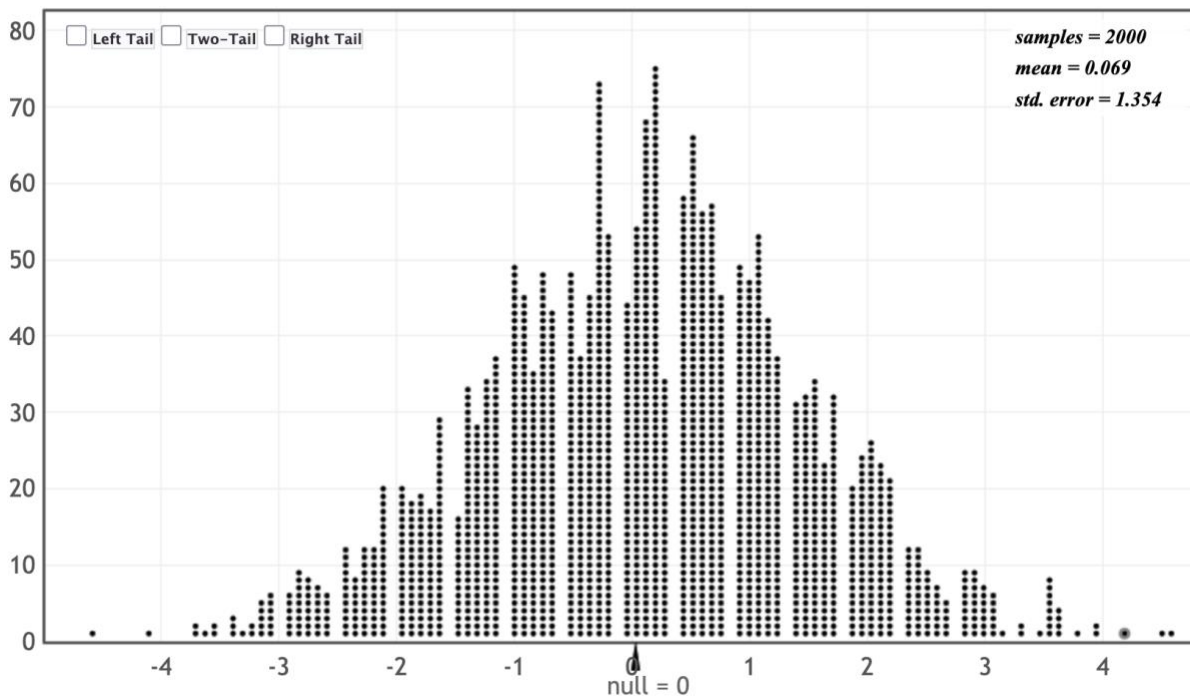
Or that beer drinkers and water drinkers attract equal numbers of mosquitoes.

$$H_a: \mu_b - \mu_w > 0$$

Or that beer drinkers attract more mosquitoes than water drinkers.

4. Use StatKey to create a randomization distribution for this test by going to the "Test for Difference in Means" link in the hypothesis test section. Notice that the bell-shaped dotplot is centered around the value 0, which represented the null hypothesis. Use the randomization distribution to indicate whether each of the following possible differences in means is (i) very likely to occur just by random chance, (ii) relatively unlikely to occur but might occur occasionally, or (iii) very unlikely to ever occur just by random chance:

Randomization Dotplot of $\bar{x}_1 - \bar{x}_2$, Null hypothesis: $\mu_1 = \mu_2$



-7 (iii)

1 (i)

-3 (ii)

-0.5 (i)

5 (iii)

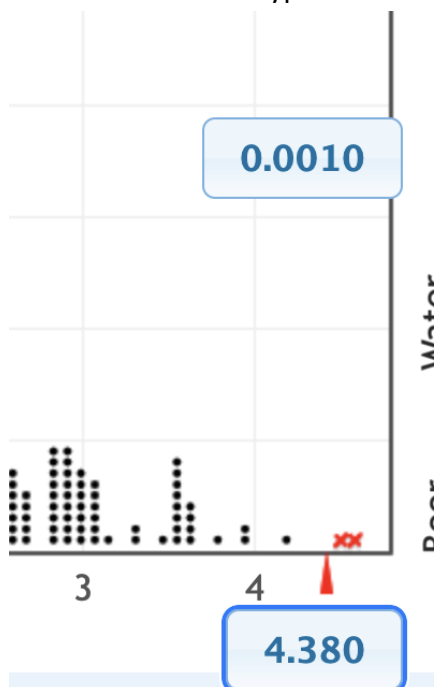
Bonus question: what does each dot on this dotplot represent?

A simulated sample, of the same size as the original experiment, created under the assumption that the null hypothesis is true.

5. Locate the actual difference in means observed in the study and shown for the Original Sample. Where does the sample statistic lie in the randomization distribution? Is it likely or unlikely to occur just by random chance?

4.38 is at the extreme right end of my graph. It is extremely unlikely to have occurred just due to random chance if the null hypothesis were true.

6. Use the sample statistic to find the p-value: click the “right tail” box, and type your sample statistic in the box that appears underneath. Is the p-value large or small? Does this provide evidence for the alternative hypothesis?



browser including Chrome, Firefox, etc. The p-value is 0.001, or one tenth of one percent. This is very small, which means this experiment is evidence to reject the null hypothesis. In other words, the two sample averages are different enough to support the idea that mosquitoes may actually be more attracted to beer drinkers.

More Hypothesis Testing: Lion Attacks after a full moon

Scientists studying lion attacks on humans in Tanzania found that 95 lion attacks happened between 6 pm and 10 pm within either five days before a full moon or five days after a full moon. Of these, 71 happened during the five days after the full moon while the other 24 happened during the five days before the full moon. Does this sample of lion attacks provide evidence that attacks are more likely after a full moon? In other words, is there evidence that attacks are not equally split between the two five-day periods?

Let p be the proportion of attacks that occur after the full moon. We want to find out if there is evidence that the true value of p is greater than 0.5.

7. What are the null and alternative hypotheses?

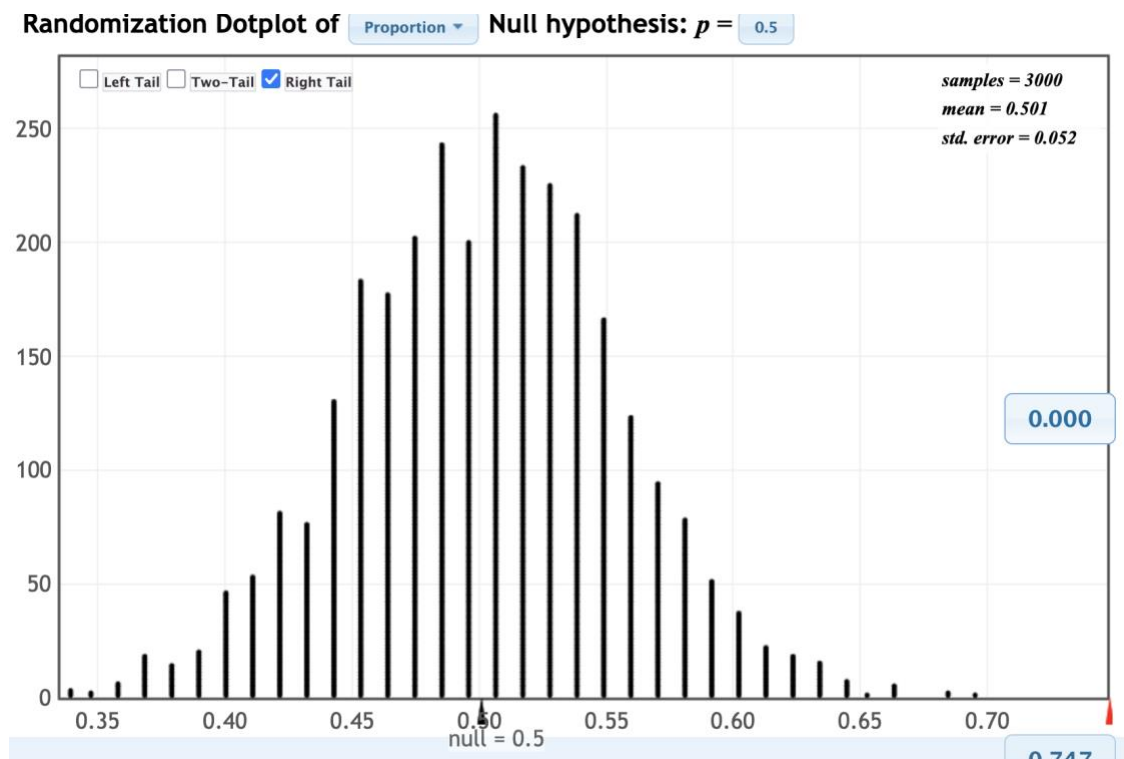
$$H_0: p = 0.5$$

Or that half of lion attacks occur after the full moon (and the other half occur before).

$$H_a: p > 0.5$$

Or that more than half of lion attacks occur after the full moon.

8. Use StatKey to find the p-value. (Note that this is a test for a single proportion since the data come from one sample.)



$p=0!$

9. Interpret the results of your hypothesis test: is this evidence that lions really do attack more after a full moon?

Since the p-value is extremely small, our sample proportion of lion attacks ($71/95$) is extremely unlikely to occur due to chance if lion attacks really happen before and after the full moon equally. This is evidence to reject the null hypothesis and support the idea that lion attacks happen more often in the five days after a full moon.